

附录 1: 参考公式

$$i_D = I_S (e^{\frac{v_D}{V_T}} - 1) \quad r_D = \frac{V_T}{I_D} \quad i_D = K_n (v_{GS} - V_{TN})^2$$

$$i_D \approx 2K_n (v_{GS} - V_{TN}) v_{DS} \quad K_n = \frac{K'_n}{2} \cdot \frac{W}{L} = \frac{\mu_n C_{ox}}{2} \left(\frac{W}{L} \right)$$

$$i_D = K_n (v_{GS} - V_{TN})^2 (1 + \lambda v_{DS}) \quad r_{ds} = [\lambda K_n (v_{GS} - V_{TN})^2]^{-1} = \frac{1}{\lambda I_D}$$

$$g_m = 2K_n (V_{GSQ} - V_{TN}) = 2\sqrt{K_n I_{DQ}} = \frac{2}{V_{TN}} \sqrt{I_{DQ} I_D} \quad R_o = R // r_{ds} // \frac{1}{g_m}$$

$$r_{be} = 200 + (1 + \beta) \frac{26(\text{mV})}{I_{EQ}(\text{mA})}$$

$$f_{L2} = \frac{g_m}{2\pi C_s} \quad f_H = \frac{1}{2\pi R'_{si} C}, \quad C = C_{gs} + (1 + g_m R'_L) C_{gd}, \quad R'_{si} = R_{si} // R_g$$

$$A_{vd1} = -\frac{1}{2} g_m (r_{ds} // R_d) \quad A_{vc1} = -\frac{g_m (r_{ds} // R_d)}{1 + g_m (2r_o)} \quad K_{CMR1} \approx g_m r_o$$

$$A_{vd1} = -\frac{\beta R_c}{2r_{be}} \quad A_{vc1} = \frac{-\beta R_c}{r_{be} + (1 + \beta) 2r_o} \quad K_{CMR1} \approx \frac{\beta r_o}{r_{be}}$$

$$R_{ic} = \frac{1}{2} [r_\pi + (1 + \beta)(2r_o)]$$

$$V_O = (1 + R_f / R_1) \left[V_{IO} + I_{IB} (R_1 // R_f - R_2) + \frac{1}{2} I_{IO} (R_1 // R_f + R_2) \right]$$

$$A_f = \frac{A}{1 + AF} \quad P_{om} = \frac{1}{2} \cdot \frac{V_{om}^2}{R_L} \approx \frac{1}{2} \cdot \frac{V_{CC}^2}{R_L} \quad P_V = \frac{2V_{CC}V_{om}}{\pi R_L} \approx \frac{2}{\pi} \cdot \frac{V_{CC}^2}{R_L}$$

$$\dot{F}_V = \frac{j\omega RC}{(1 - \omega^2 R^2 C^2) + j3\omega RC} \quad \dot{F}_V = \frac{1}{3 + j \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)}$$

$$V_L = (1.1 \sim 1.2) V_2 \quad I_L = \frac{0.9V_2}{R_L}$$